Lecture 5: Trading Volume Alpha

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Introduction

Volume prediction from statistical perspective

Portfolio strategy based on volume prediction

Investment performance evaluation

Motivation

AI/ML for investments

- Powerful tools for portfolio optimization
- Superiority in extracting trading signals that forecast future returns

Challenges: ML for Finance

- From prediction to implementation
  - Trading costs
  - A wedge between academic and practitioner research
- From statistical loss to economic loss

Key Message

- Beyond return forecasting
- Economic value

- Portfolio optimization chiefly focuses on return and risk prediction
- In implementation: trading costs
  - With trading costs, a key trade-off:
    - Passive vs. aggressive trading, depending on trading cost forecasts
  - Price impact depends on how much to trade, how much others are trading ...
- This paper: focus on trading volume
  - Forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha"

Why trading volume?

- Individual trade sizes are hard to obtain
- Market liquidity proxied by volume, unknown ex ante
- A general yet critical component of trading costs modeling
- Kyle (1985); Frazzini, Israel, and Moskowitz (2018):

$$PriceImpact_{a,i,t} \propto ParticipationRate_{a,i,t} := \frac{\$Traded_{a,i,t}}{\$Volume_{i,t}}$$

Volume is generic across investors (a), exogenous, observable

- Volume forecast informs trading decision: Volume $\uparrow$ $\Rightarrow$ Price impact $\downarrow$ $\Rightarrow$ Trade more aggressively (vice versa)

Main Contributions

- Forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha"
- New dataset, neural network methods, and new benchmarks
- Transfer statistical learning into finance problem solving
- Substantial economic gain, comparable to return prediction

→ **Volume prediction from statistical perspective**

- Portfolio implementation based on volume prediction
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Daily stock-level dollar trading volume

Data structure

- Unit of observation: stock, day (i, t) panel
- Sample period from 2018 to 2022, 5 years or 1,200+ days

  3-year train + 2-year testing
  4,700+ stocks in total, avg 3,500 observed stocks per day
  4,400,000+ observations in total

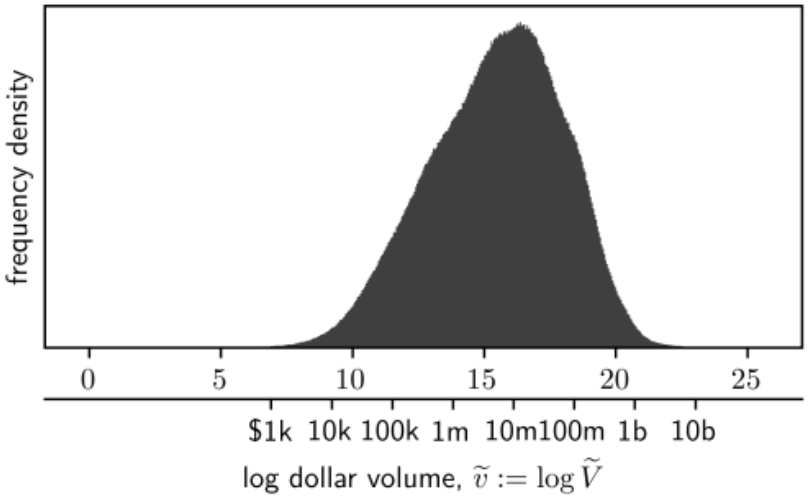
Predictors

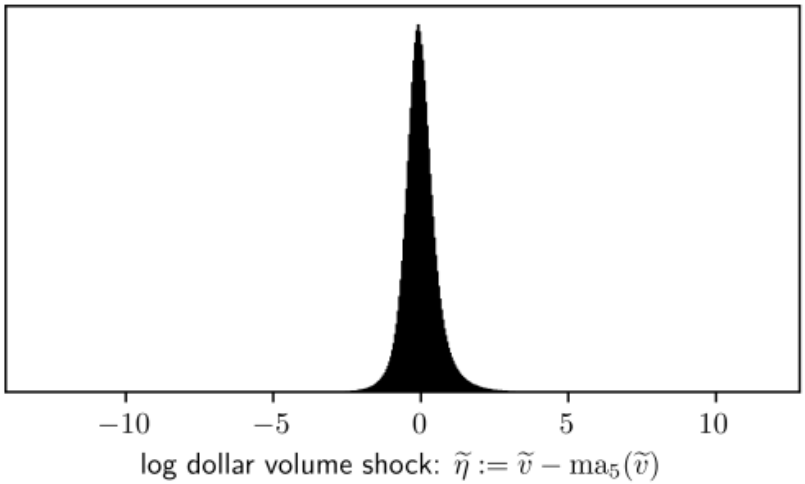
Predictors, $X_{i,t}$, 175 in total, 4 categories:

- Technical signals:
  return and volume lagged moving averages
- Firm characteristics:
  size, b/m, ..., developed for return prediction (JKP dataset)
- Deterministic calendar events:
  triple witching[1], index rebalancing, early closing etc.
- Scheduled earnings announcements

[1] Triple witching is the quarterly expiration of stock options, stock index futures, and stock index options contracts all on the same day.

Dollar volume, log dollar volume, and its shock





- Dollar volume $\tilde{V}$; log dollar volume $\tilde{v}$
- Take ma_5 as baseline throughout the paper:
  predict log dollar volume shock $\tilde{\eta} := \tilde{v} - ma_5$

  Like predicting return := log(price) − log(yesterday's price)

Neural network tools

Run (OLS and NN) regressions of $\tilde{\eta}_{i,t}$ on $X_{i,t}$

Transparent and standard machine learning implementation

- nn (neural network):
  3-layer, fully-connected, 32–16–8 ReLU nodes
- rnn (recurrent neural network):
  to capture time-series predictive dependency

$$(\hat{\eta}_{i,t}, \text{state}_{i,t}) \;=\; \mathrm{rnn}\big(X_{i,t}, \text{state}_{i,t-1}\big)$$

architecture: LSTM as the first hidden layer, all else the same

Prediction results, OOS R^2 (%)

cumulative	tech	fund-1	fund-2	calendar	earnings
# of predictors	8	14	161	165	175
A: R^2 relative to $\widetilde{\eta}$ $(\widetilde{v} - \mathrm{ma}_5)$					
ma_5	0				
ols	12.09	12.26	12.27	14.85	15.99
nn	14.31	14.90	14.42	17.13	18.45
rnn	15.80	16.25	15.47	18.12	19.86
B: number of parameters					
ma_5	0				
ols	9	15	162	166	176
nn	961	1,153	5,857	5,985	6,305
rnn	6,049	6,817	25,633	26,145	27,425

- Volume prediction from statistical perspective
  $\rightarrow$ **Portfolio strategy based on volume prediction**
- Investment performance evaluation

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Portfolio optimization problem

Trade-off: track a target portfolio vs. avoid trading cost:

Given starting \$ position x^0, target \$ position x^*, choose new position x or trade amount $\Delta x := x - x^0$ (for each stock each day)

$$\min_{\{x_{i,t}\}} \sum_{i,t} \left(\mathrm{TrackError}_{i,t} + \mathrm{TradeCost}_{i,t}\right)$$

Portfolio optimization problem

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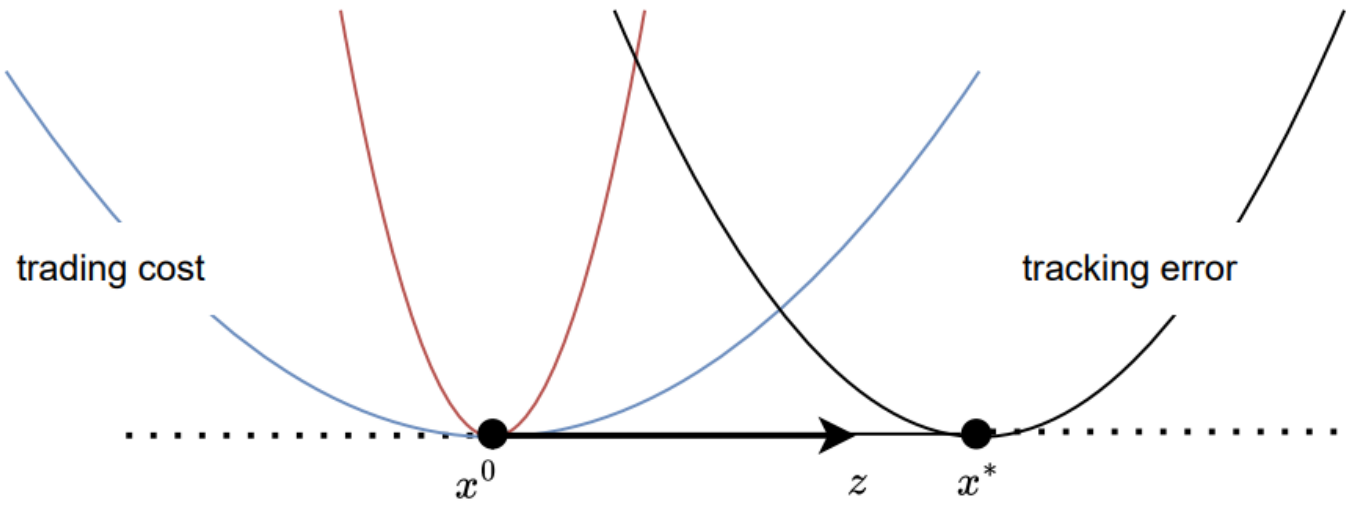

- Choose $z \in [0, 1]$ (stay put vs. trade all the way)
- Predicting volume well benefits this optimization

  Intuition: trade more aggressively if believe $\tilde{v}$ high, vice versa

Tracking error modeling

$$\text{TrackError} := \frac{1}{2}\mu(x^* - x)^2$$

- **Microfoundation**: from a mean-variance optimization problem, x^* embeds return signals, AUM, leverage, etc.
- μ controls the penalty for being away from the target (like a hyperparameter). Larger $\mu \Rightarrow$ trade more aggressively.

Trading cost modeling

$$TradeCost := \frac{1}{2}\,\tilde{\lambda}\,(x - x^0)^2 \qquad \tilde{\lambda} := \frac{0.2}{\tilde{V}}$$

Microfoundation: *PriceImpact* linear in participation rate:

$$PriceImpact = 0.1\,\frac{x - x^0}{\tilde{V}}, \qquad TradeCost = PriceImpact \cdot (x - x^0)$$

E.g., buying 10% of daily volume incurs 1% price impact (Frazzini, Israel, and Moskowitz, 2018)

Combined objective

Minimize the expectation of *TradeCost* + *TrackError* conditional on information

$$\min_x \ \mathbb{E}\left[\frac{1}{2}\,\tilde{\lambda}\,(x - x^0)^2 + \frac{1}{2}\,\mu\,(x^* - x)^2 \,\middle|\, \text{conditioning information}\right]$$

Importantly, *TradeCost* is unknown ex ante when choosing x: $\tilde{\lambda}$ depends on $\tilde{V}$.

Combined objective

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Importantly, TradeCost is unknown ex ante when choosing x: $\tilde{\lambda}$ depends on $\tilde{V}$.

Key point: predicting $\tilde{V}$ well helps solving this problem.
Intuition: more adaptive to liquidity condition:
trade more aggressively if believe $\tilde{V}$ high, patiently otherwise.

Next: analyze and solve the problem to demonstrate this point and show the economic gains.

Objective function analysis

Isolate target trade size:

Define trading rate $z := \dfrac{x - x^0}{x^* - x^0}$, then the target becomes

$$\begin{aligned} &\tfrac{1}{2}\tilde{\lambda}(x - x^0)^2 + \tfrac{1}{2}\mu(x^* - x)^2 \\ &= \tfrac{1}{2}(x^* - x^0)^2 \underbrace{\left(\tilde{\lambda} z^2 + \mu(1 - z)^2\right)}_{\mathrm{loss}_{\mathrm{econ}}(\tilde{v}, z; \mu)} \end{aligned}$$

New problem: minimize expected economic loss by choosing z

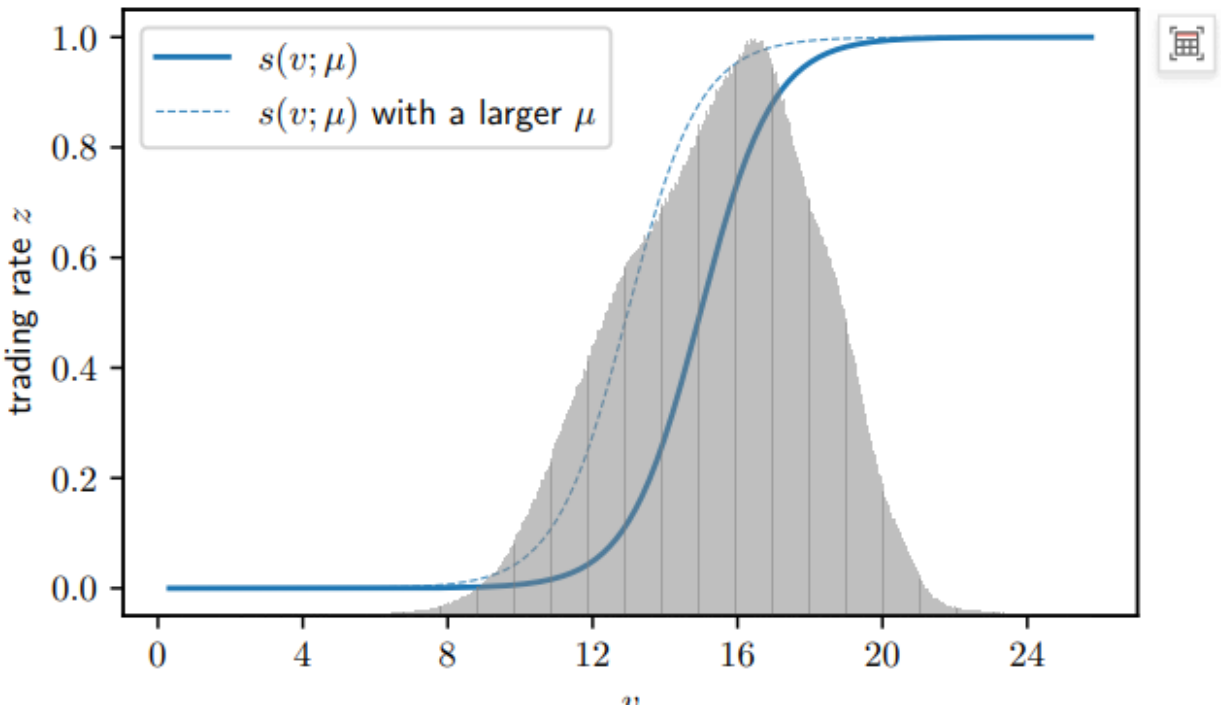
$$\min_z \ \mathbb{E}[\mathrm{loss}^{\mathrm{econ}}(\tilde{v}, z; \mu) \mid X]$$

- same problem regardless of tracking target (will experiment with different targets)
- for now, choose trade rate $z \in [0, 1]$, solve the normalized problem

$\text{loss}^{\text{econ}}$ analysis: z and v relationship

say we believe $\tilde{v} = v$, that is $\lambda = 0.2/\exp(v)$:

$$\arg\min_z \ \text{loss}^{\text{econ}} = \frac{\mu}{\mu + \lambda} = \text{sigmoid}\big(v - \log(0.2/\mu)\big) \longrightarrow s(v; \mu)$$



- A forecast of v corresponds to a choice z.

  Conversely, a choice z implies an (underlying) forecast of v.

Approach 1, statistical learning

plug-in stats forecast $\hat{v}$, trade $\hat{z} = s(\hat{v}; \mu)$

1. ML based on a statistical loss, MSE:

$$\min_{v(\cdot)} \sum_{i,\, t \in \text{train}} \text{loss}^{\text{least squares}}(\tilde{v}_{i,t},\ v(X_{i,t}))$$

2. Plug $\hat{v}_{i,t} = v(X_{i,t})$ into the optimal policy $s(\cdot; \mu)$
   Choose trading rate $\hat{z}_{i,t} = s(\hat{v}_{i,t}; \mu)$

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- **Evaluated by economic loss**

$$\frac{1}{|\text{test}|} \sum_{i,t \in \text{test}} loss^{\text{econ}}\big(\tilde{v}_{i,t}, \hat{z}_{i,t}; \mu\big)$$

$$\Big[\ loss^{\text{least squares}}(\tilde{v}, v) = (\tilde{v} - v)^2, \quad loss^{\text{econ}} = \ldots \text{ as defined}\ \Big]$$

Approach 2, economic learning

directly learn z to optimize economic loss

$$\min_{z(\cdot)} \sum_{i,\, t \in \text{train}} \text{loss}^{\text{econ}}(\tilde{v}_{i,t},\ z(X_{i,t}))$$

Approach 2, economic learning:
directly learn z to optimize economic loss

$$\min_{z(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{econ}}\big(\tilde{v}_{i,t}, z(X_{i,t})\big)$$

Equivalently, this is also predicting volume,
because choosing z implies a forecasted value v

$$\min_{v(\cdot)} \sum_{i,t \in \text{train}} loss^{\text{econ}}_{vv}\big(\tilde{v}_{i,t}, v(X_{i,t})\big)$$

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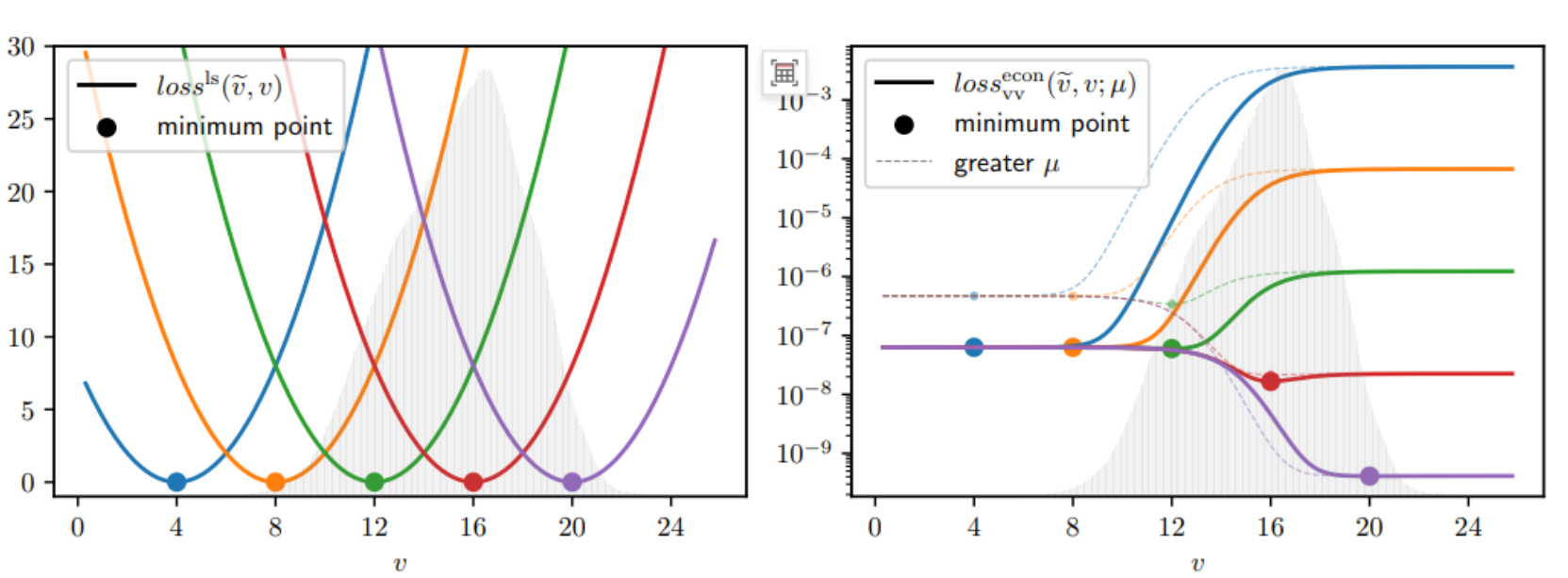
Comparing the two methods:

- Only difference is at the loss functions

Next, loss functions comparison and implications on learning

Comparing statistical and economic loss



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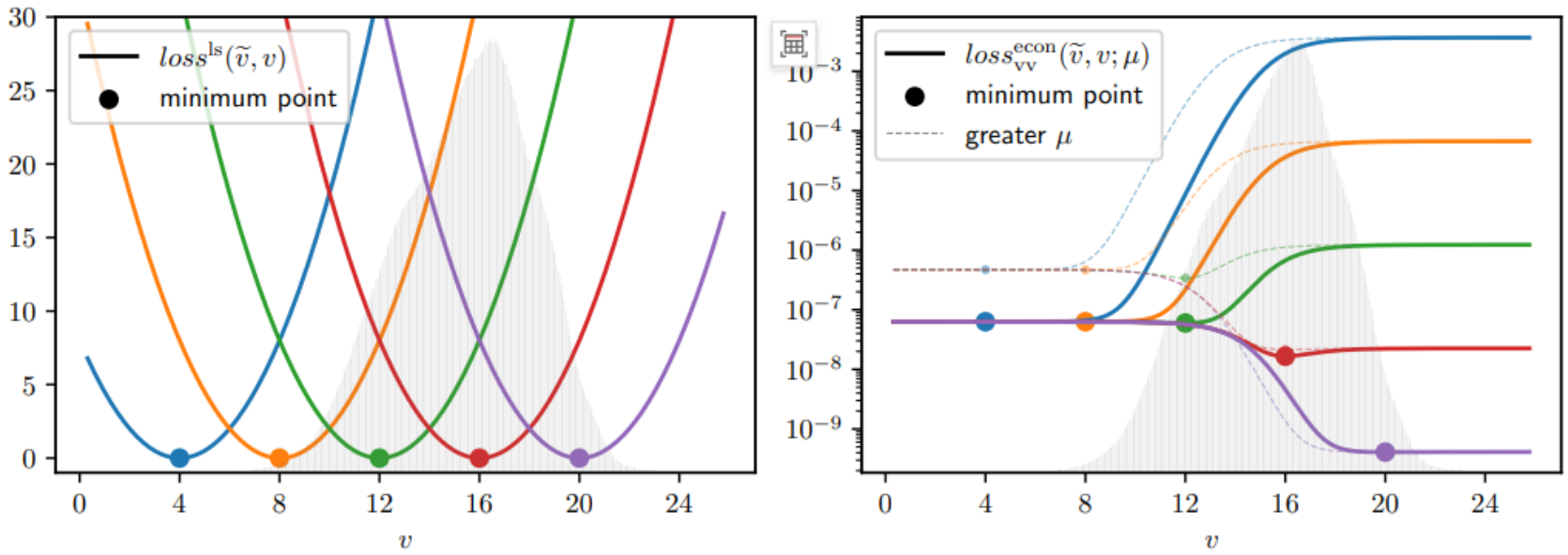


Economic loss is asymmetrical

- huge cost if actual $\tilde{v}$ is low but predicted as high
  $\rightarrow$ trade a lot when liquidity is low
- in contrast, opportunity cost for not trading is bounded

Economic objective favors conservatism to avoid overestimating volume

Theoretical analysis

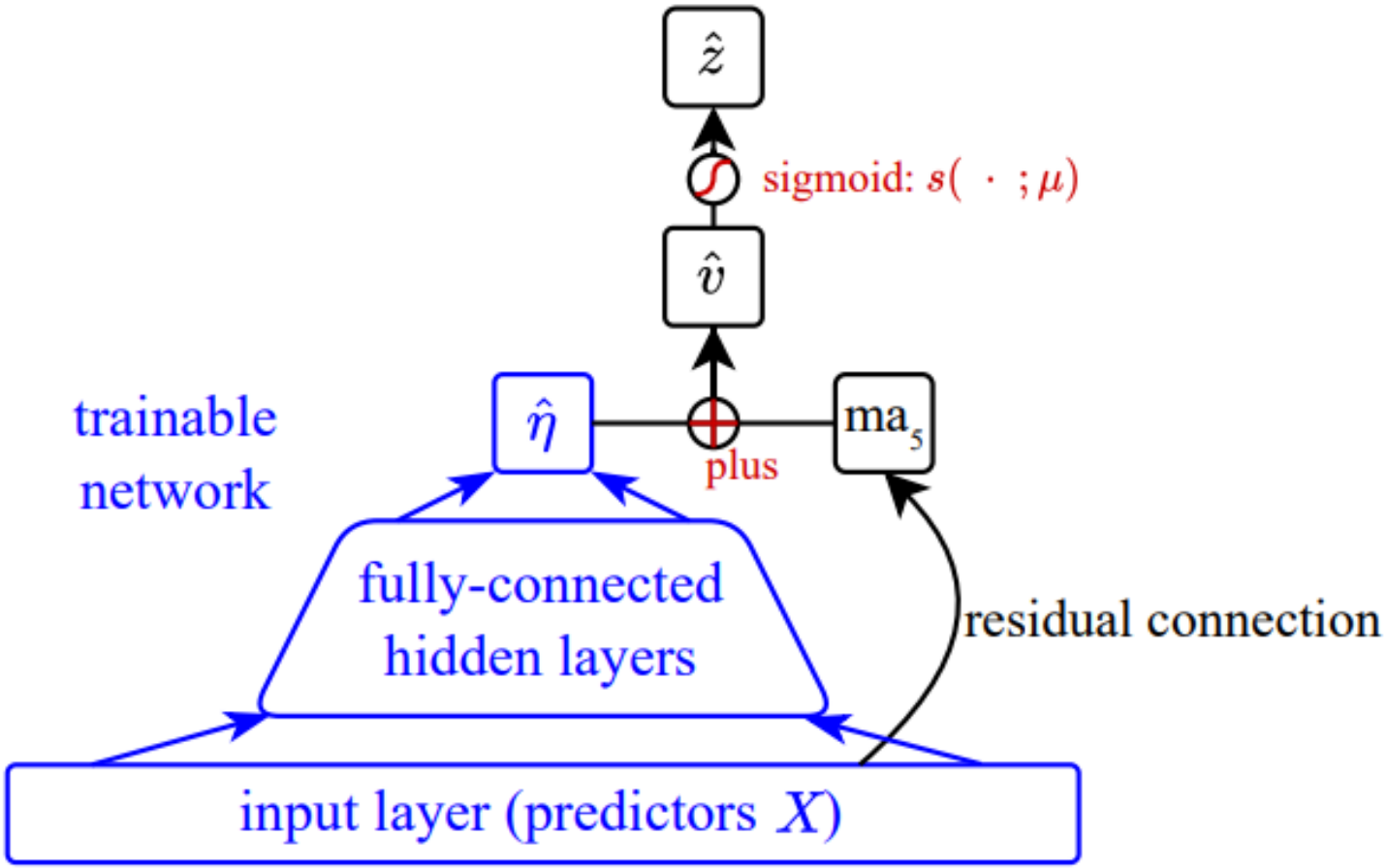
Proposition: least squares optimizer does not maximize economic objective, even in population

- well-known: conditional expectation is the least squares minimizer

$$\mathbb{E}[\tilde{v} \mid X] = \min_{v \in \sigma(X)} \mathbb{E}\Big[\text{loss}^{\text{least squares}}(\tilde{v}, v)\Big]$$

- how about other loss functions?
  iff the loss function is in the Bregman class (Banerjee, Guo, and Wang, 2005; Patton, 2020)
  however, $\text{loss}^{\text{econ}}_{vv}$ is not in Bregman class

Network architecture, transfer learning



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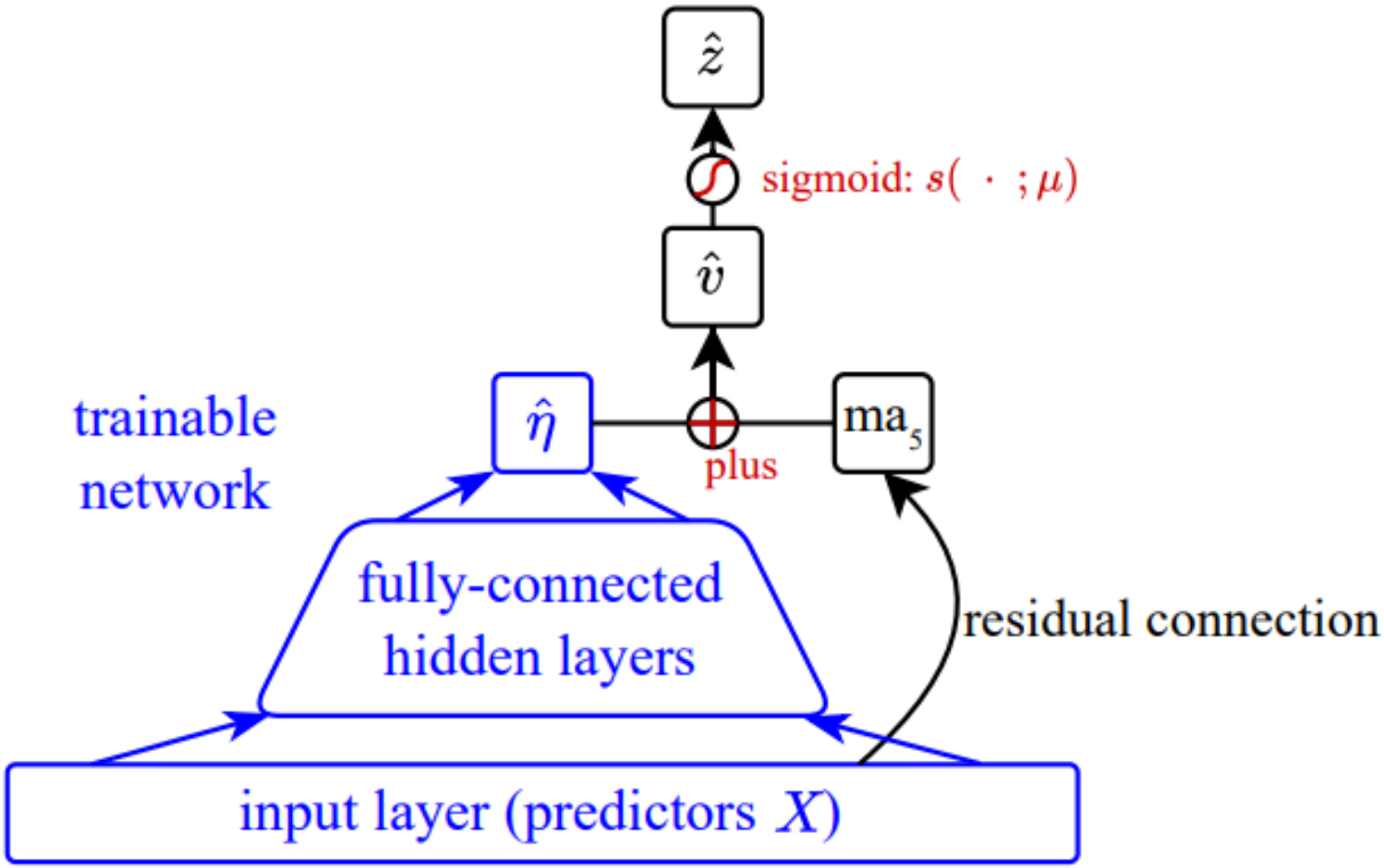

Transfer learning:

- Statistical learning fits an NN model, applicable to all μ values and serves as a pre-trained foundation model.
- Economic learning fine-tunes the NN for each μ for downstream tasks, since $loss^{\text{econ}}$ is indexed by μ.

Transfer learning, summary

- Method 1, statistical learning:
  - run ML regressions of $\tilde{v}$ onto X, learn $v^{\text{nn}}(X)$ in training sample
  $$\min_{\theta} \sum_{i,\, t \in \text{train}} (\tilde{v}_{i,t} - v^{\text{nn}}(X_{i,t}; \theta))^2$$
  - plug-in and trade $s(v^{\text{nn}}(X_{i,t}; \theta^*), \mu)$ in testing sample

- Method 2, economic learning:
  - directly learn z:
  $$\min_{\theta} \sum_{i,\, t \in \text{train}} \text{loss}^{\text{econ}}(\tilde{v}_{i,t},\, z^{\text{nn}}(X_{i,t}; \theta); \mu)$$
  - trade $z^{\text{nn}}(X_{i,t}; \theta^*_{\mu})$ in testing sample

Method 1 fits one nn model, applicable to all μ values
Method 2 fine-tunes the nn for each μ, because $\text{loss}^{\text{econ}}$ is indexed by μ

- Volume prediction from statistical perspective
- Portfolio strategy based on volume prediction

  $\rightarrow$ **Investment performance evaluation**

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Trading experiments, setup

Apply the strategies ($\hat{z}$) to track various sets of trading targets $\{x^*_{i,t}\}$ that mimic different trading tasks, including

1. simulated quantitative trading strategy implementing *before-cost* high-Sharpe strategies
2. factor zoo portfolios (JKP factors) implementing different investment styles

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1. simulated quantitative trading strategy
   implementing ***before-cost*** high-Sharpe strategies
2. factor zoo portfolios (JKP factors)
   implementing different investment styles

Experiment procedure: recursively track the supplied target

- given $x^0_{i,t}$, $x^*_{i,t}$, trade $x_{i,t} = x^0_{i,t} + \hat{z}_{i,t}(x^*_{i,t} - x^0_{i,t})$
- roll to $x^0_{i,t+1} := x_{i,t}(1 + r^f_t + r_{i,t})$
- trade $x_{i,t+1}$ to track $x^*_{i,t+1}$, and so on ...

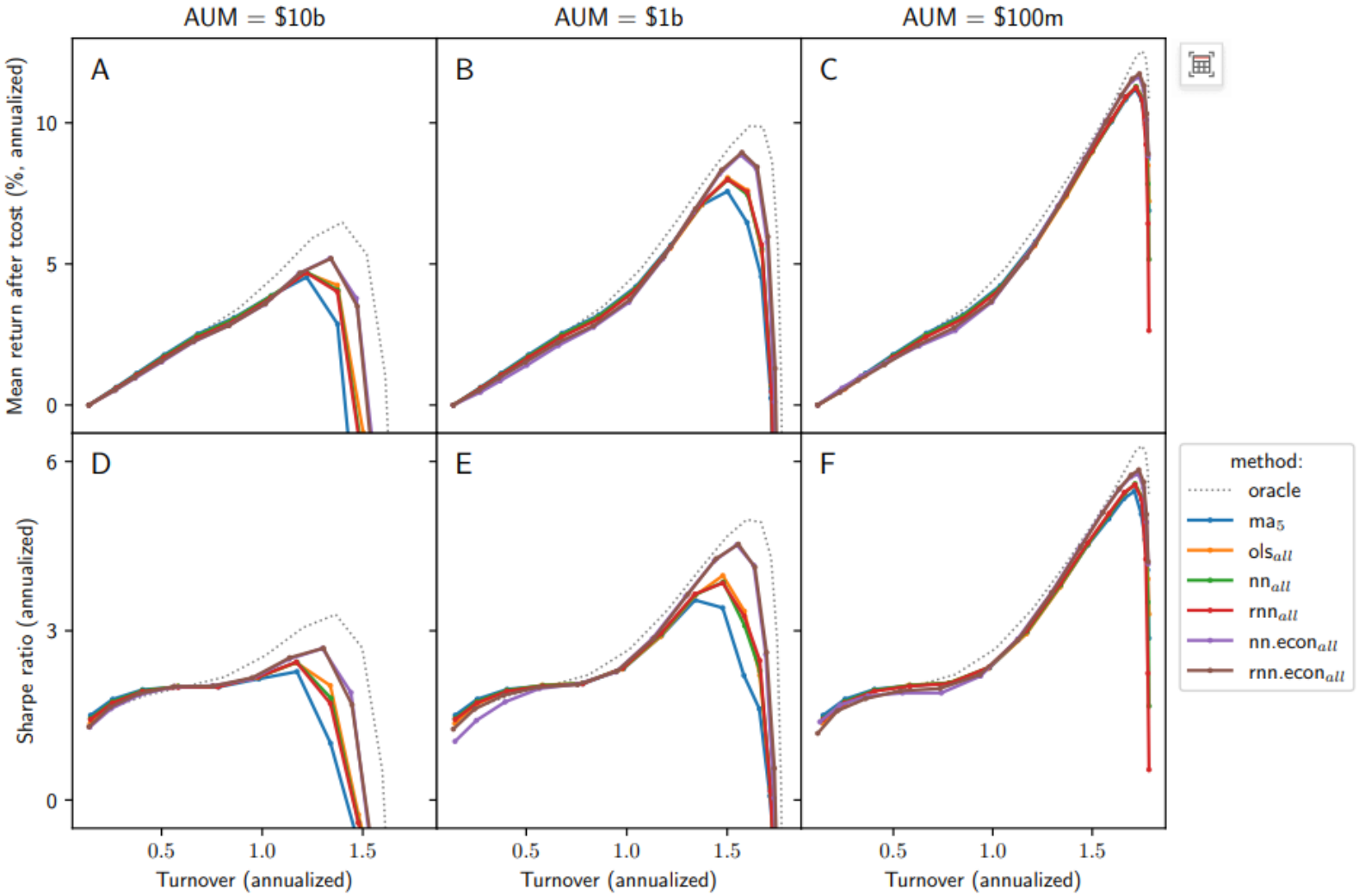
Evaluate the investment outcome of $\{x_{i,t}\}$ in the testing sample

Experiment 1, quantitative trading task

Construct targets $\{x^*_{i,t}\}$ with unrealistically high (before-cost) investment performance

- Sharpe ratio ≈ 7
- blow up to various AUM magnitudes

Performance across μ, all 175 predictors



Performance with μ hyper-parameter tuning

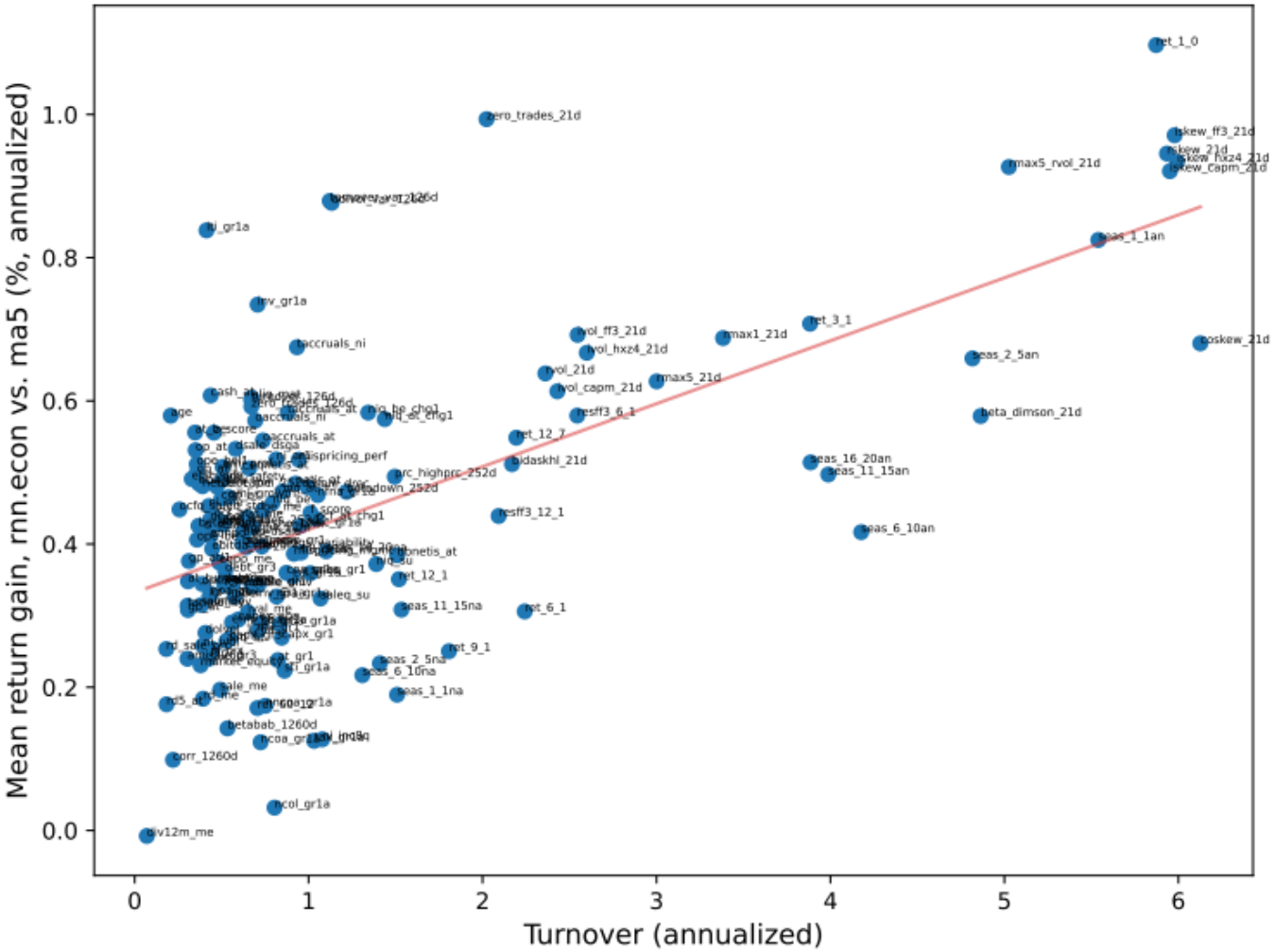
A. Mean return (%, annualized)					B. Sharpe ratio (annualized)			
AUM	\$10b	\$1b	\$100m	\$10m	\$10b	\$1b	\$100m	\$10m
ma_5	3.88	6.47	11.19	13.20	2.00	2.21	5.47	6.55
ols_{tech}	3.82	7.60	11.28	13.14	2.16	3.32	5.59	6.59
nn_{tech}	3.76	7.30	11.32	13.13	2.14	2.79	5.63	6.59
rnn_{tech}	3.74	7.84	11.33	13.13	2.18	3.58	5.64	6.59
nn.econ_{tech}	4.60	7.20	11.29	13.22	2.13	2.57	5.59	6.63
rnn.econ_{tech}	4.67	8.69	11.59	13.30	2.50	4.32	5.73	6.66
ols_{all}	3.82	7.60	11.28	13.14	2.17	3.35	5.60	6.59
nn_{all}	3.86	7.44	11.28	13.13	2.19	3.09	5.60	6.58
rnn_{all}	3.79	7.55	11.25	13.09	2.18	3.26	5.59	6.56
nn.econ_{all}	4.64	8.87	11.61	13.29	2.18	4.24	5.74	6.66
rnn.econ_{all}	4.68	8.95	11.77	13.30	2.50	4.53	5.85	6.68
oracle	6.47	9.89	12.54	13.56	3.05	4.97	6.28	6.80

Experiment 2, implementing factor zoo portfolios

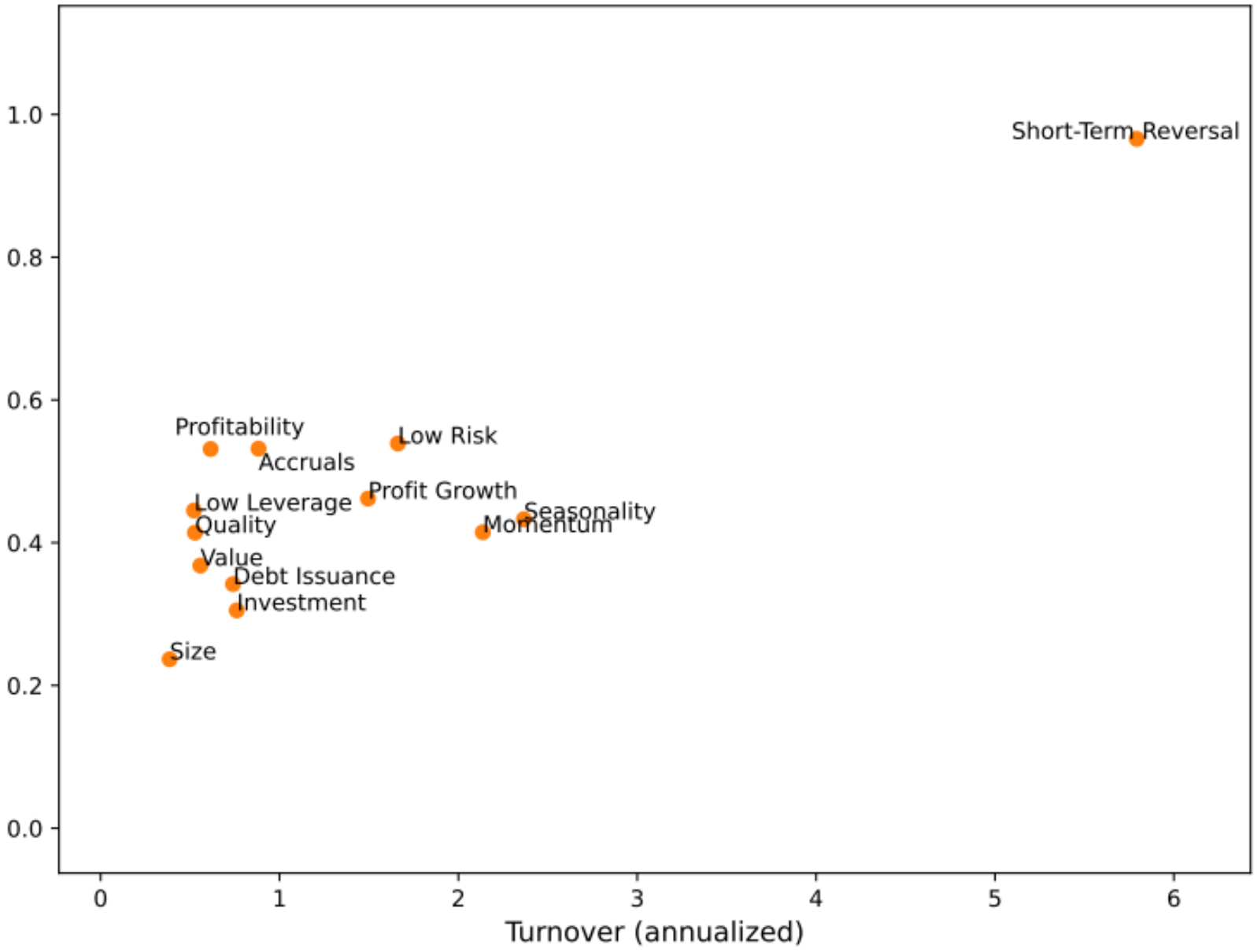
Long-short sorted portfolios for each 153 JKP characteristics

- AUM $10b, leverage 100%

Mean return improvements in implementing factor portfolio



Mean return improvements averaged by theme clusters



Trading volume alpha, summary

- daily firm-level trading volume is highly predictable, especially with machine learning
- forecast volume $\rightarrow$ portfolio strategy $\rightarrow$ net-of-cost "alpha" economic value comparable to return prediction
- transfer learning
  from statistical prediction to economically motivated trading tasks

Economically important frontier in portfolio research

References

Banerjee, A., X. Guo, and H. Wang. 2005. On the optimality of conditional expectation as a Bregman predictor. *IEEE Transactions on Information Theory* 51:2664–2669. Conference Name: IEEE Transactions on Information Theory.

Frazzini, A., R. Israel, and T. J. Moskowitz. 2018. Trading Costs. doi:10.2139/ssrn.3229719.

Kyle, A. S. 1985. Continuous auctions and insider trading. *Econometrica* 53:1315–1335. Publisher: [Wiley, Econometric Society].

Patton, A. J. 2020. Comparing Possibly Misspecified Forecasts. *Journal of Business & Economic Statistics* 38:796–809. Publisher: Taylor & Francis. doi:10.1080/07350015.2019.1585256.

Microfoundation of trading model

Quadratic tracking error as the result of mean–variance optimization

$$A(1+r_t^f) + \sum_i x_{i,t}\, m_{i,t} - \frac{\gamma}{2A} \sum_{i,j} x_{i,t}\, x_{j,t}\, \sigma_{ij,t}^2 - \sum_i \mathrm{TradingCost}_{i,t}$$

$$\Rightarrow\ -\tfrac{\gamma\sigma^2}{2A} \textstyle\sum_i \left(x_{i,t} - \underbrace{\frac{A}{\gamma\sigma^2}\, m_{i,t}}_{x_{i,t}^*} \right)^2 - \textstyle\sum_i \mathrm{TradingCost}_{i,t} + \left(A(1+r_t^f) + \tfrac{A}{2\gamma\sigma^2} \textstyle\sum_i m_{i,t}^2 \right)$$

(assuming homoscedasticity $\mathrm{Var}[r_{i,t+1}] = \sigma^2$ *and zero covariances)*